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AIPI 590: Alternative Data

Fall 2026

Dr. Pramod Singh

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Class attendance



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Welcome to our third lecture!
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Teams for the Term Project

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Team members/Individuals	Number
Colby Robbins	1
Haydn Stucker	2
Qingyi Hu, Shenwei Zhang	3
Keming Zhou	4
Snehal Jindal	5
Moonwon Seo (Online) , Celine Yang (Online)	6
Kailong Zhang	7
Debshree Chowdhury, George Deng	8
Hadley McCormack, Konur Nordberg	9
Xianyu Wang	10
Supriya Nannapaneni	11
Burak Donbekci	12

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Teams for the Thurro Project

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Team	Problem Statement
Haydn Stucker Colby Robbins Keming Zhou	EV production and ancillary investment
Shenwei Zhang Konur Nordberg Qingyi Hu	India high-frequency consumption nowcast
Snehal Jindal Supriya Nannapaneni Hadley McCormack	MNREGA wage spikes and rural FMCG
Celine Yang Moonwon Seo Kailong Zhang	Renewable grid-connectivity backlog
Donbekci, Burak Wang, Xianyu Chen, Xinlei Deng, George	Residential launch-to-sales velocity

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Where we are

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Week	Date	Topic
Week 1	Aug 26, 2026	Foundations of Alternative Data
Week 2	Sep 2, 2026	Types of Alternative Data
Week 3	Sep 9, 2026	The AI Lifecycle in Alternative Data Products
Week 4	Sep 16, 2026	Presentation of Ideas for the Term Project
Week 5	Sep 23, 2026	Investment Research Introduction
Week 6	Sep 30, 2026	Investment Research with Alternative Data – Methods & Pitfalls
Week 7	Oct 7, 2026	Investment Research with Alternative Data – From Models to Strategy
Week 8	Oct 14, 2026	Mid-Semester Presentations
Week 9	Oct 21, 2026	Ethics and Explainability of Alternative Data Products
Week 10	Oct 28, 2026	Case Study Debate & Alternative Data Simulation
Week 11	Nov 4, 2026	Synthetic Data in Alternative Data - Expanding the Edge
Week 12	Nov 11, 2026	Industry Project Presentation
Week 13	Nov 18, 2026	Term Project Presentations

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Recap from Last Class

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Duke Alt Data Interest Form

WHAT THE FORM ASKS

- Who you are
- Which project or track interests you
- Hours per week you can sustain: 3–5, 6–9, 10–14, or 15+
- What you already know, and what you want to learn



Scan to apply

Or email altdatalabcenter@duke.edu and we will send it to you.

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Industry Project - Five projects with Thurro

01

Consumption nowcast



High frequency

02

Rural FMCG signal



Granularity + text

03

Grid backlog → IPP risk



Non-obvious link

04

EV → investment siting



Leading indicator

05

Real estate → cement



Aggregation method

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Did you connect to Thurro's MCP?

1

Connect

Add the Thurro connector in Claude (Pro access) or Chat GPT plus. Each team needs at least one member with access — if nobody on a team has it, tell your TA and we'll sort it out.

2

Ask in plain English

No SQL required to get started — ask for a dataset or schema the way you'd ask a colleague. Claude translates the question into a query against Thurro.

3

Get structured answers

Claude returns the data — tables, fields, coverage — ready to pull into Python, R, or a notebook for your own analysis.

ILLUSTRATIVE EXAMPLE

Student: "What's the schema for the UPI transactions dataset?"

Granularity: state, month · Fields: transaction volume, transaction value · History: 3+ years · Update frequency: monthly

Related tables you may also want: vehicle registrations (state, monthly), GST collections (state, monthly)

Thurro Contact Details –

Please remember to copy TA team on all communications

For office hours, reach out to:

Akhilesh Tilotia

Co-Founder, Thurro
akhilesh@thurro.com

Ayush Gandhi

AVP – Investment Research, Thurro
ayush.gandhi@thurro.com

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Structured Intelligence

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Term project - Key Milestones & Deliverables

5 %	15 %	20 %
Idea Pitch	**Mid-Semester Review**	**Final Demo**
Sep 16 (Next Week!)	Oct 14	Nov 18

Deliverables (all decks due before class on the due date)

- **Idea Pitch deck**
- **Mid-Semester deck**
- **Final deck**
- **5-min demo video**
- **GitHub repo**

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Geolocation as Alternative Data

Foot Traffic + Card Spend

Links visits to spending

Store-level conversion; brand/ticker performance

Geolocation Trends

Migration hotspots & population shifts

Real estate & housing-demand forecasts

Footfall Density over Time

Best sites for new-store expansion

Seasonal & event-driven surges

Satellite as Alternative Data

Military Activity: troop movements, base expansions, ship locations

Environmental Monitoring: methane emissions, deforestation, climate impact

Retail Analysis: parking-lot traffic to predict big-retailer sales

Infrastructure & Construction: new projects, urban growth, illegal mining

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Placer.ai Discussion

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Discussion Prompts -

- **Which brand's growth strategy impressed you the most? What about it stood out?**
- **How do different patterns (short visits, suburban families, office commuters) affect outcomes for chains?**
- **Can coffee chains keep expanding without hurting existing stores? Why or why not?**
- **What other industries could benefit from similar location-based insights?**
- **What might this data miss about consumer behavior?**

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Podcast Discussion

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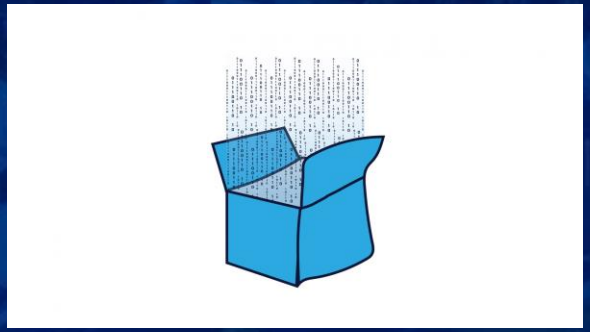
Podcast: All About Alternative Data

Some discussion topics -

- How the sector grew up, and why COVID was the moment location and card data proved themselves.
- How a fund decides a dataset is worth paying for, which is the checklist from the vetting slide.
- What AI and large language models are doing to the industry you are about to walk into.

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From Alternate Data to Data Products



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Converting Alternate Data to Data Products

The Challenge

Raw Alternative Data is Messy

The Formula

Data Product = Data + Models + Delivery

The Ultimate Goal

Usable Business Insights

Value Transformation Examples

Satellite Raw Signal
“Retail Traffic Index”

Geolocation Data
“Footfall Heatmap

Credit Card Transactions
“Consumer Spending Trend”

Social Media Activity
“Sentiment Score”

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The AI Product Reality Check – The Theranos of AI



The Story: Amazon's "Just Walk Out" (2018-2024)

Just Walk Out



Amazon's cashier-less checkout technology

What Everyone Saw....



Amazon launched **"Just Walk Out"** in 2018 - a cashier-less checkout technology that let shoppers grab items and leave without scanning anything.

Positioned as **the future of retail**, eliminating checkout lines entirely.

Media excitement about a CV/AI breakthrough.

Demo videos showing shoppers grabbing items and walking straight out.

The tech required massive human oversight - not the "just AI" system it was marketed as

Human reviewers were **required to watch checkout footage,** but shoppers didn't know humans were watching

No more than **20-50 reviews per 1,000 sales** was Amazon's internal target

Controversary, **700** reviews per 1,000 sales were needed - by over 1,000 human reviewers

What the PM Actually Dealt With.....

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April 2024 — Repositioned, Not Killed

BUSINESS

Amazon removes 'Just Walk Out' cashier-less tech from Fresh grocery stores

PUBLISHED WED, APR 3 2024 · 2:10 PM EST

CNN Business Reporting by CNN Staff

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The 2025 Example: Builder.ai

2016 – 2025

“Building an app should be as easy as ordering pizza.”

Peak valuation ~$1.5bn. $230m raised in 2023 from investors including Microsoft and the Qatar Investment Authority. Fast Company’s #3 most innovative company in AI.

Insolvent, 20 May 2025.

Sources: Wall Street Journal (Aug 2019); Bloomberg (May 2025); Reuters and the Financial Times on the insolvency filing.

What Everyone Saw....



Builder.ai promised app-building "as easy as ordering pizza," run by an AI named Natasha.

Pitched as the AI that replaces your dev agency.

Fast Company's #3 most innovative AI company, 2023.

$230m raised in 2023 — backers included Microsoft.

The Wall Street Journal reported in 2019 that humans, not AI, did most of the coding.

Customers were never told where Natasha ended and an engineer began.

Bloomberg: revenue round-tripped with VerSe, 2021–2024.

Insolvent 20 May 2025. ~80% of staff gone by June.

What the PM Actually Dealt With.....

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Builder.ai Timeline

- 2016 Founded. An AI assistant, Natasha, will build your app for you.
- Aug 2019 The Wall Street Journal reports that human engineers — not AI — do most of the coding.
- 2023 Raises $230m (Microsoft, Qatar Investment Authority). ~$1.5bn valuation. Fast Company's #3 most innovative company in AI.
- 27 Feb 2025 Founder-CEO Sachin Dev Duggal resigns.
- May 2025 Bloomberg reports revenue round-tripped with VerSe, 2021–2024. Insolvency announced 20 May after creditors seize $50m.
- Jun 2025 ~1,000 staff — about 80% of the company — gone.
- **The WSJ story ran six years before the collapse. The information was public, and the market priced it at zero.**

So Where Is the Line?

- Human-in-the-loop is not fraud. Amazon labelling video after the fact to train a model is ordinary supervised learning. Engineers behind Natasha was, by itself, a product decision.
- **The line is disclosure and in 2025 it was drawn in court.**
- Nate, an “AI-powered” one-click shopping app: in April 2025 the US Department of Justice charged founder Albert Saniger with fraud. Checkout was substantially performed by human contractors in the Philippines.
- Amazon’s engineering was defensible. Its disclosure was not. Builder.ai’s use of engineers was defensible. Its revenue reporting was not.
- **For your term project: if someone acted on your data product because of how you described it, would the description survive a look under the waterline?**

That is what we call the ***AI Product Manager's Dilemma,*** and it applies directly to data products.

The AI Product Manager's Dilemma



When you manage an AI or data product, you are caught between groups:

Engineers / Data Scientists

- **"Model has 94% accuracy!"**
- **But accuracy in lab success in real-world use.**
- **Example: Geolocation data may work in cities but fail in rural areas.**

PM's job Ask: *"Does this solve real user problems?"*

Business Leaders / Executives

- **"We spent $2M — where's ROI?" "Competitor launched AI last month, we need it now!"**
- **Expect instant results from data products.**

PM's job Explain: *"AI/data takes longer than traditional features - cleaning, validation, adoption."*

End Users

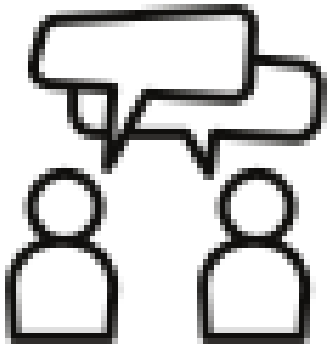
- “Why this weird recommendation?"
- They don’t care about algorithms → only want reliable results.
- **Example:** Credit card data says *“demand is up”* but stores look empty → trust collapses.
- **PM’s job** → Build trust, explainability, confidence indicators.

At this point, we see the core problem:

- Engineers speak the language of accuracy,
- Business speaks the language of ROI,
- Users speak the language of trust.

So how do we, as product managers, bring all three together?

Learn frameworks to navigate these conversations successfully



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Frameworks for Managing Data Products



Managing Data Products

Essential frameworks for data adoption and design.

Note:

These are some of the most widely used frameworks, but they are not the only ones. Different organizations adapt or combine frameworks depending on context.

01 **CRISP-DM**
Data product lifecycle

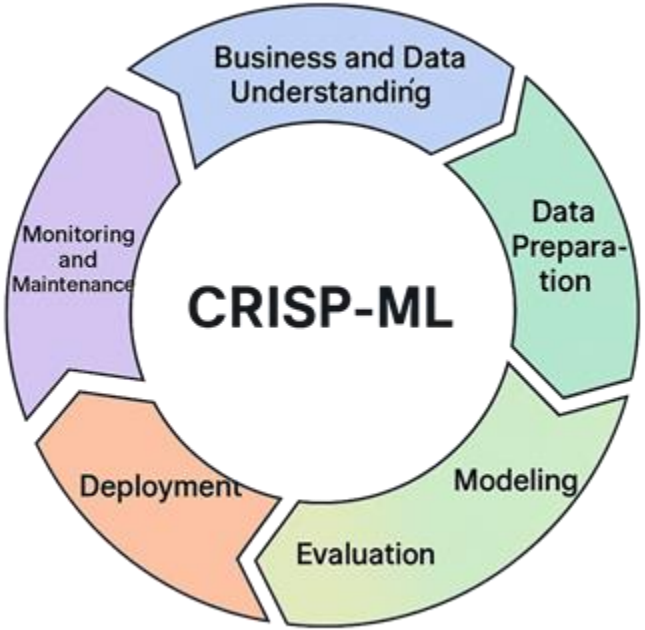
02 **Google PAIR**
Human-centered data design

03 **BCG 10-20-70**
Adoption matters more than algorithms

04 **Gartner AI Maturity Model**
Journey of data adoption

05 **AI Ethics**
Trust, fairness, regulation

The CRISP-ML Methodology for the Lifecycle for Data Products



The Industry Standard: Cross-Industry Standard Process for Data Mining (CRISP-DM)

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The Six Phases

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1 **Business Understanding -** Why use this dataset? (e.g., satellite → retail footfall)



Engineering Focus:

Define technical requirements



Business Focus:

ROI and success metrics



User Focus:

What problem are we solving?



PM Role:

Ensure all three align on the same goal

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❷ **Data Understanding**: Is it usable? Complete? Biased?



Engineering :

"Do we have enough quality data?"



Business Focus:

"How much will data collection cost?"



User Focus:

"How will our data be used?"



PM Communication: Translate data needs into business investment

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3 **Data Preparation** - Cleaning, feature engineering, integration



Engineering :

"Data cleaning and feature engineering"



Business Focus:

"Why does this take 40% of the timeline?"



User Focus:

No direct impact (behind the scenes)



PM :

Set expectations that this phase takes the longest.

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4 **Data Modeling** - Build predictive models using data



Engineering :

"Algorithm selection and training"



Business Focus:

"When will we see results?"



User Focus:

Still no visible progress



PM Communication: "Foundation work for the user experience"

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5 **Evaluation** - Does it align with business KPIs?



Engineering :

"Model performance metrics"



Business Focus:

"Does this meet our success criteria?"



User Focus:

"How will this actually help me?"



PM role:

Connect technical performance to business/user value

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6 **Deployment** - Deliver insights via dashboards, APIs, or apps.



Engineering :

"Production integration and monitoring"



Business Focus:

"Time to measure ROI"



User Focus:

“Finally see the AI feature in action"



PM role:

User adoption and feedback loops

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From Lifecycle to User Design

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- CRISP-DM helps us **manage the data lifecycle** (from business understanding to deployment).
- But **deployment is not the finish line,** people must trust **and use** the product.
- This brings us to the next framework: **Human-Centered Data Product Design (Google PAIR)**

Google's PAIR (People + AI Research) Guidelines



The Established Framework: Human-Centered Data Product Design

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The Four Pair Principles we will focus on

The Four PAIR Principles

01

Understand User Context

Who is using the data? What decisions are they making?

02

Show Confidence Levels

Don’t just give a prediction, also show probability and uncertainty.

03

Enable User Control

Let users adjust filters, override, or explore the underlying data.

04

Plan for Failure

Anticipate missing data, system errors, or unexpected outputs.

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Understand User Context
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Understand User Context

Who is the user?

Analyst, PM, trader, store manager, exec?

Decision window

Real-time, daily, weekly, quarterly?

Risk tolerance

What's an acceptable false positive/negative?

Data literacy

How much statistical detail can they handle?

Environment

Mobile vs desktop; low-connectivity cases; accessibility.

Success metric

Tie outputs to a user KPI (e.g., *increase conversion +15%*, *reduce stockouts*)

Examples: Aligning Data to User Needs

Satellite data
Retail Ops

Store managers require practical, high-level guidance to make day-to-day decisions.

Target Outcome:

Operational actions (e.g., labor scheduling)

Geolocation data
Mall Traffic

Analysts need structured, filterable datasets to dissect performance across dimensions.

Target Outcome:

Segmentable trends (by hour or store cluster)

Credit Card data
Spending Index

Investors seek broad market direction paired with statistical guardrails.

Target Outcome:

Macro signals with coverage caveats

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Show Confidence Levels

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Show Confidence Levels: Key Guidelines

01
Display Uncertainty
Use prediction intervals, error bars, or High–Med–Low confidence badges.

02
Sample Size & Coverage
Represent sample size (n) and % of population, merchants, or zip codes.

03
Data Freshness
State "Last updated 2h ago" along with general data source recency.

04
Quality Indicators
Track and display key metrics like latency, missing-rate, and coverage changes.

05
Explain "Why"
Provide top drivers or attributions (e.g., demand up due to weekend event).

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Enable User Control

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User Control Capabilities

Filtering & Segmentation

Time window, location, store cluster, shopper cohort.

Thresholds & Alerts

User-set alert rules (e.g., notify if footfall drops >10% WoW).

Scenario Controls

Sliders and toggles for assumptions (weather, promos, events).

Explainability on Demand

“Why this?” expands to show drivers; link to a model card.

Export & API

Equipped with privacy and aggregation safeguards.

Examples of User Control

Satellite

Toggle **cloud-masked only** to filter out low-quality imagery.

Pick **weekday vs weekend** to analyze behavioral shifts.

Geolocation

Filter **residents vs visitors** to isolate core customer behavior.

Use radius sliders (**250m–1km**) to adjust spatial granularity.

Credit Card

Select target **merchant categories** to refine commercial trends.

Exclude statistical **refund spikes** to clean baseline spend data.


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Plan for Failure Modes
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Operationalizing Model Robustness

Data Issues

Anticipate missing days, vendor outages, schema changes, and seasonal anomalies.

Degradation

Fall back to heuristics or baselines, and gray out affected features with clear labels.

Guardrails

Monitor with drift checks, quality dashboards, SLA alerts, and a bad model kill switch.

Compliance

Ensure privacy using aggregation thresholds, differential privacy, and audit logs.

Examples: Mitigation in Action

Satellite

Heavy cloud cover

Action: Revert to historical trend + confidence downgrade.

Geolocation

iOS policy change

Action: Coverage drop banner + recalibrated indices.

Credit Card

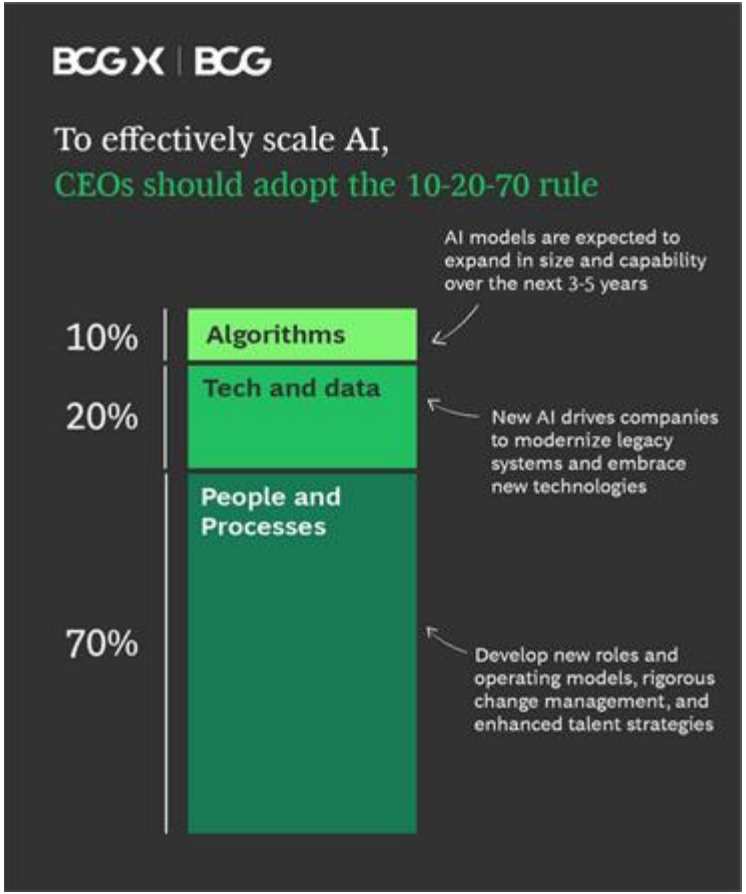
Vendor latency

Action: Show **status badge** + last reliable date.


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From Usability to Adoption
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- PAIR helps us make **data products usable and trustworthy**.
- But usability alone is not enough. Products must also be **adopted across the organization**.
- Success often depends less on algorithms and more on **people, processes, and change management**.
- Next: The **10-20-70 rule**.

BCG's AI Product Development Framework (10-20-70)



- **10% Algorithms** → Model accuracy & performance
- **20% Tech & Data**→ Pipelines, computing resources, integration
- **70% People & Processes** → Adoption, workflow integration, change management
- **Key insight:** Data product success = **less about tech, more about adoption**

The 10-20-70 Rule:

10%

Algorithms

- **What it means:** The actual AI/ML algorithms and models.
- **Engineering focus:** Model accuracy and performance.
- **PM insight:** Algorithm choice is the smallest success factor.

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The 10-20-70 Rule:

20%

Data and Technology Infrastructure

- **What it means:** Data pipelines, computing resources, integration platforms.
- **Business focus:** Platform investment and technical foundation.
- **PM insight:** Technology and data provide the foundation for AI success.

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The 10-20-70 Rule:

70%

People, Processes, and Change Management

- **What it means:** User adoption, workflow integration, organizational transformation.
- **User focus:** How AI fits into daily workflows and culture.
- **PM insight:** Success depends on people and processes, not technical factors.

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From Adoption to Readiness

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- BCG showed us: adoption matters more than algorithms.
- But adoption also depends on **organizational readiness**.
- Not every company is at the same stage of using AI or alternative data.
- Next: **Gartner's AI Maturity Model**

Gartner

Gartner's AI Maturity Model

The Industry Standard: Gartner's AI Maturity Assessment Framework

The Five Levels of AI & Data Maturity

01
Awareness

Knows AI and alt data exist. No live projects.

e.g. *A retailer curious about geolocation data — not buying it.*

Tell: Can name three vendors. Has bought from none.

02
Active

Pilots and PoCs in isolated teams. ROI unclear.

e.g. *A bank piloting card data to predict spending patterns.*

Tell: A working notebook with no owner.

03
Operational

In production for specific functions. Impact is measurable.

e.g. *A fund trading on card data; a chain siting stores by footfall.*

Tell: Someone's bonus now depends on it.

04
Systemic

Integrated across processes. Enterprise strategy and governance.

e.g. *A retailer blending satellite, social and card data to forecast demand.*

Tell: Governance exists — and it can say no.

05
Transformational

Data reshapes the business model. New revenue from data itself.

e.g. *A payments network selling aggregated spending insights.*

Tell: Data sits on the revenue line, not the cost line.

Take the company you interned at or any company you follow closely. Where is it on this ladder, and what is the **ONE** thing that would move it up a level?

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From Maturity to Responsibility

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- Reaching higher maturity levels means **AI/data are everywhere** in the organization.
- But more power = more responsibility.
- Without **ethics, transparency, and fairness**, adoption breaks down.
- Next: **AI Ethics Framework**

AI ETHICS FRAMEWORK

The Responsibility Standard: Partnership on AI Ethical Guidelines for Product & Data Development

5 Most Common AI Ethics Principles



Source: Gartner
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Gartner

Fairness & Non - Discrimination

- AI/data must not discriminate against protected groups
- Test for bias across demographics & geographies
- Ensure equitable outcomes for users
- Example: LinkedIn checking job recommendations for gender bias

Accountability & Transparency

- Organizations must be responsible for AI/data decisions
- Provide audit trails and decision logs
- Clear ownership and escalation paths
- Example: Credit scoring models that explain why someone was rejected

Explainability & Interpretability

- Users should understand how outputs are generated
- Provide user-friendly explanations, not just technical metrics
- Use model interpretability tools when needed
- Example: Medical AI showing which symptoms led to diagnosis

Explainable AI in Practice

GradCAM — explaining vision models (CNNs)

- **What it means:** Produces a heatmap over an image showing which pixels most drove the model's prediction
- **Example:** A pneumonia classifier's heatmap lights up the lung opacities — not the hospital ID stamp in the corner of the X-ray

Random Forest — explaining tabular models

- **What it means:** Feature importance ranks which input variables reduce prediction error the most, averaged across every tree in the forest
- **Example:** A credit-risk model ranks “missed payments in the last 6 months” above zip code — supporting a compliant, defensible decision

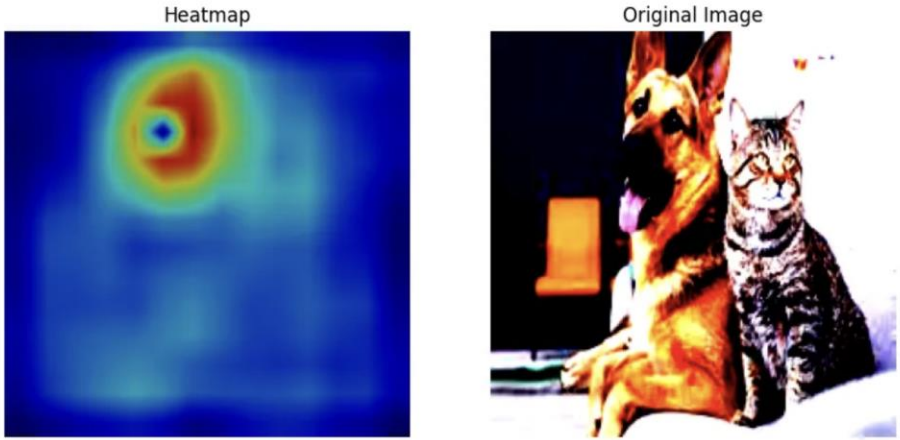
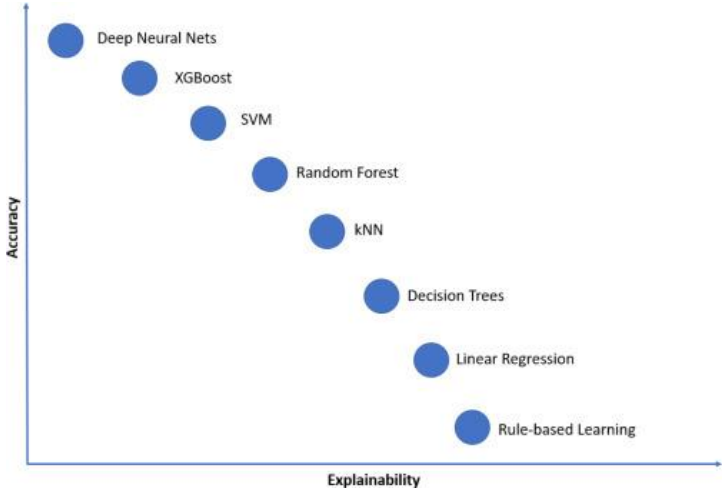


fig 1: activation map for dog



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CLASS ACTIVITY / EXPLAINABILITY

Ban ice cream. Save swimmers?

A beach safety model uses ice cream sales to predict shark attacks.
The town proposes banning ice cream.



Would you approve the ban?

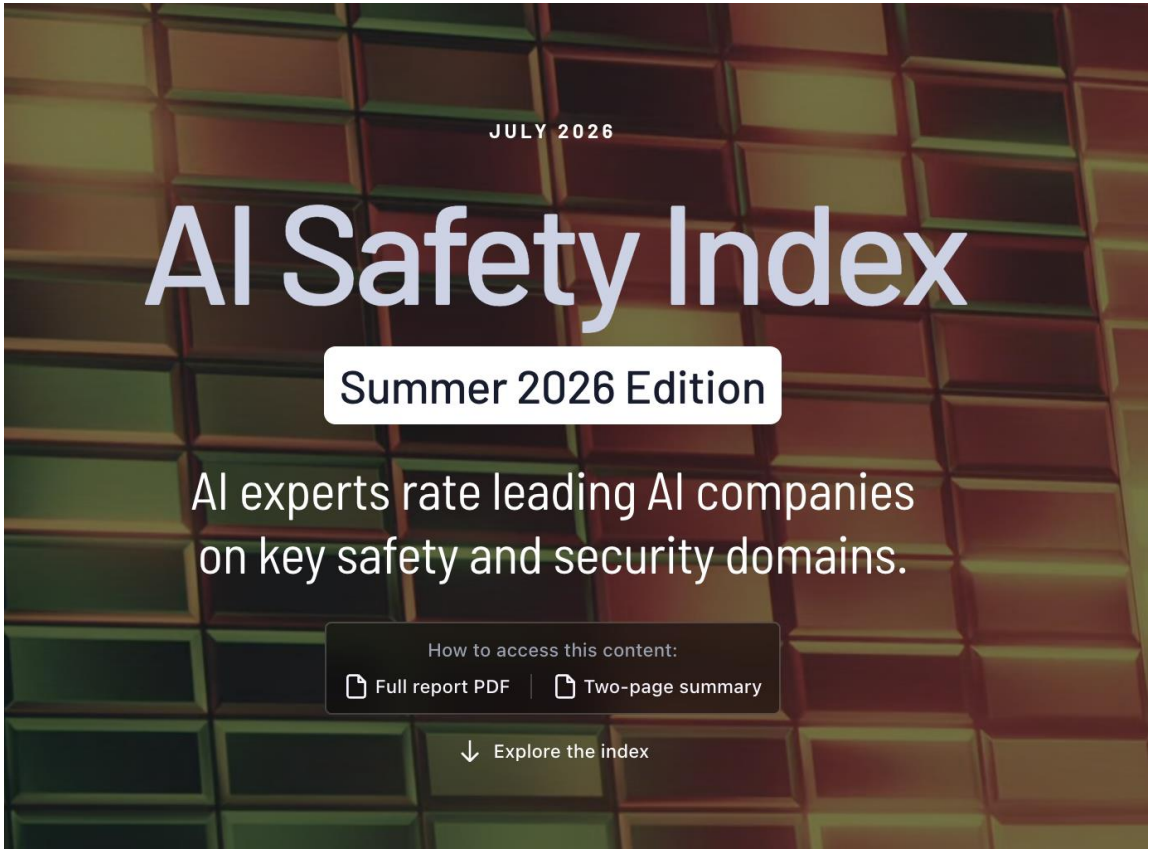
APPROVE / REJECT / NEED MORE EVIDENCE

1. What else could explain the link?

2. What extra data would you check?

3. How would you explain the prediction?

Discuss in pairs. Share one better explanation.



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	Anthropic	OpenAI	Google DeepMind	Meta	Z.ai	Alibaba Cloud	xAI	DeepSeek	Mistral
Overall Grade	C+	C	C	D+	D-	D-	F	F	F
Score	2.66	2.28	2.01	1.32	0.88	0.87	0.65	0.47	0.33
Grade Trend (Winter 25)	C+	C+ ▼	C	D ▲	D ▼	D-	D ▼	D ▼	N/A
Domains Hint: Click on a domain to inspect									
Risk Assessment 6 indicators	C+	C+	C+	D+	F	F	D-	F	F
Current Harms 9 indicators	B-	C	C	D-	C-	C-	F	D-	F
Safety Frameworks 4 indicators	B-	C+	C	C-	D-	D-	D	F	F
Existential Safety 4 indicators	D+	D+	D	F	F	F	F	F	F
Governance & Accountability 4 indicators	B	C	C-	D+	F	D-	F	F	F
Information Sharing 10 indicators	B+	B-	B-	D+	D	D	D	D-	D-
Survey Responses	✓	✓	✓	✓	✓	–	–	–	–

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Privacy & Security

- Protect user data at every stage
- Apply anonymization, aggregation, and privacy-preserving methods
- Comply with regulations (GDPR, CCPA, etc.)
- Example: Aggregated mobility data instead of individual device traces

The Question the Other Four Can’t Answer

WHAT we COVERED

- Fairness
- Accountability & Transparency
- Explainability
- Privacy & Security

FIFTH : It asks whether you should build it at all.

CONSENT

Did the people in this dataset agree to be in it — and would they, if it were explained to them plainly?

BENEFIT

Who is better off, by name? If the only beneficiary is the buyer, say that out loud rather than dressing it up.

SCALE

What happens when this works perfectly and every competitor has it too? Is that world better or worse?

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Bringing it all Together

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Summary of Core Principles

Key takeaways and frameworks for AI product management

Alternative Data

Messy raw signals must become **products**

CRISP-ML

Standard model **process discipline**

PAIR

Google toolkit **human-centered design**

BCG 10-20-70

AI scaling rule **adoption > algorithms**

Gartner Maturity

Assessment **know your stage, plan next step**

Ethics

Foundation **trust, fairness, sustainability**

AI product management is not about models, but about creating usable, trusted, and adopted data products


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The Alternative Data Pipeline
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Where and how do you source Alt data?

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REST APIs & SDKs

Endpoint

A URL that returns structured data.
GET https://api.stlouisfed.org/fred/series/observations
?series_id=UNRATE&api_key=...

Auth

Most alt data APIs use API keys (header or query param). OAuth 2.0 for user-scoped data (e.g., LinkedIn). Bearer tokens for rate-limited tiers.

Pagination

Bulk datasets are paginated — never assume one request = full dataset. Use offset, cursor, or page params to iterate.

Rate Limits

Free tiers: 100–500 req/day. Paid tiers: thousands/min. Always cache responses locally — don't re-hit APIs you've already pulled.

Real APIs our Students Use

FRED (St. Louis Fed)

800k+ US + international time series 100+ sources CPI, unemployment, interest rates. Python: fredapi library.

Nasdaq Data Link

Formerly Quandl. Alt data + economic time series. pandas-datareader compatible.

Google Trends API

Unofficial via pytrends. Search volume normalized 0–100 by keyword.

SEC EDGAR (EDGAR API)

Full-text filings search since 2001. 10-K/Q, 8-K, insider trades via JSON API.

Alpha Vantage

Fundamental data, earnings, news sentiment, economic indicators.

How a Quant Fund Sources Satellite Data

Used by funds like Two Sigma and Citadel.

Data Vendor Contract

Sign a data license with Planet Labs or Maxar Technologies. Coverage: 200+ satellite captures/day globally. Typical cost: $50k–$500k/yr for institutional license.

Raw Imagery Delivery

Daily S3 bucket drops of GeoTIFF files. Each image is tagged with lat/lon, timestamp, cloud cover %, and satellite ID. ~10–50GB per day depending on coverage area.

Computer Vision Pipeline

CNN or object detection model (YOLO-style) counts cars in retail parking lots, oil tanks, container ships. Models retrained quarterly as imagery quality improves.

Signal Normalization

Raw car counts normalized by lot size, weather (rainy days suppress parking), and day-of-week. Adjusted for construction or lot closures via change detection.

Alpha Generation

Berkeley Haas (2024): satellite parking data generated 4–5% abnormal returns in 3 days around retail earnings. Signal is strongest 2 weeks before the announcement.

Data Cleaning for Alt Data

Common Dirty Data Problems

Missing values

Satellite imagery has cloud cover gaps. Credit card panels under-represent cash users. Google Trends returns '<1' for rare queries.

The fix : Interpolation for time series; flag-and-impute for sparse panels; rolling median fill for sensor dropout.

Timestamp misalignment

Job postings scraped at 11pm EST are stored as next-day UTC. FRED releases data with 3-week lags. Satellite captures vary ±2hrs by orbit.

The fix : Normalize all timestamps to UTC at ingest. Always store as-of date + release date separately.

Duplicates

Web scrapers re-crawl the same listing. API pagination overlaps. A job posted on Indeed AND LinkedIn appears twice.

The fix : Deduplicate on (source, entity_id, date). For fuzzy text: MinHash or Jaccard similarity at 0.85 threshold.

Scale inconsistency

Google Trends returns 0–100 relative scores — not absolute volumes. Foot traffic counts vary by panel size across vendors.

The fix : Normalize within-series using z-score or min-max. Never compare raw Trends numbers across keywords.

Activity : Spot the Problem

1. Foot Traffic Data

Store	Date	Visits	Timestamp
Target #042	2024-11-28	0	2024-11-29 06:00 UTC
Target #042	2024-11-28	847	2024-11-28 23:00 EST
Walmart #107	2024-11-28	null	2024-11-28
Walmart #107	2024-11-29	1,203	2024-11-29

2. Job Posting API Response

{
title: "Senior ML Engineer"
company: "Stripe"
salary: "competitive"
posted_date: 2024-09-01
source_1: Indeed
source_2: LinkedIn
source_3: Stripe.com/careers
job_id_1: IND-8823
job_id_2: LIN-99210
job_id_3: STR-442
status: open
last_seen: 2025-01-14
}

Feature Engineering on Alt Data

Temporal Features

Rolling window aggregates

e.g. 7-day, 30-day, 90-day avg foot traffic → smooths sensor noise

Lag features

e.g. Lag-1 / lag-4 of weekly card spend → captures momentum and mean-reversion

YoY / QoQ % change

e.g. Same-store job count YoY → removes seasonal hiring cycles

Day-of-week & seasonality

e.g. Monday foot traffic / monthly avg → removes weekday bias in retail data

Cross-Sectional Features

Percentile rank vs peers

e.g. Company's card spend rank within retail sector this week

Market share proxy

e.g. Firm's foot traffic / total sector foot traffic from Placer.ai

Relative sentiment score

e.g. Company NLP score minus industry median — removes macro noise

Surprise vs consensus

e.g. Observed card spend − analyst estimate → standardized surprise

NLP & Text Features

Sentiment polarity

e.g. VADER / FinBERT on earnings call transcripts → tone shift detection

Topic prevalence

e.g. LDA topic weight for 'supply chain' in 8-K filings → risk flag

Linguistic uncertainty

e.g. Count of 'approximately', 'may', 'could' in CEO commentary

Job posting skill density

e.g. TF-IDF weight of 'LLM', 'RAG', 'agent' in JD text → AI investment signal

Enriching: Combining Alt Data Signals

Signal Stacking

Each alt data source becomes a feature in a shared model. XGBoost trained on foot traffic + card spend + job postings simultaneously. Most common approach at multi-strategy funds.

Ensemble / Model Blending

Separate model per source, outputs combined via weighted average or meta-learner. Revelio Labs job model + Bloomberg Second Measure card model → blend at signal layer.

Knowledge Graph Enrichment

Alt data nodes linked via a graph (company → supplier → customer). Port traffic signal enriched with company supply chain graph to predict downstream revenue impact.

Now let's take a look at how AI Product Management helps us package these data signals into real products

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Build vs Buy

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What Are You Actually Buying?

01
Acquisition

Panels, satellites, licences, collection infrastructure — and the legal right to hold it.

Buy. Almost always.

02
Cleaning

Dedup, timestamp normalisation, gap-filling, outlier handling.

Buy. It is most of the fee.

03
Entity mapping

Company name → ticker → parent. The stage nobody advertises and everybody needs.

Buy, then verify.

04
Enrichment & features

Your indices, your normalisation, your surprise-vs-consensus. Same feed, different signal.

BUILD. Your alpha is here.

05
Delivery

A CSV drop, an API with an SLA, or a dashboard. Same data, very different price.

Depends on your user.

A satellite licence runs **$50k–$500k a year**. Two or three engineers to rebuild stage 01 costs more than that; before you have a single usable row.

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Buy vs. Build: What Each One Costs You

BUY

Licence someone else's feed

What you get

Coverage and history you cannot recreate · cleaning and entity mapping already done · an SLA and someone to call · signal in weeks, not quarters.

What it costs you

$50k–$500k a year, renewed · lock-in the moment your pipeline assumes their schema · their coverage gaps and their biases become yours · and no differentiation — your competitor bought the same feed.

BUILD

Collect and process it yourself

What you get

Exclusivity and a defensible asset · full control of coverage, methodology and history · no renewal, no vendor risk, no rebrand mid-contract.

What it costs you

Two to three engineers and 12–18 months before usable signal · you own the consent, privacy and legal risk · you own maintenance forever · and it usually costs more than the licence.

Before You Sign: Five Questions

01

COVERAGE

What fraction of the universe, and where are the holes? Ask for the holes, not the coverage number.

02

HISTORY

How far back — and was it collected then, or backfilled later? Backfilled history has look-ahead baked in.

03

POINT-IN-TIME

Do I get as-of date and release date separately? If not, you cannot backtest honestly.

04

ENTITY MAPPING

Who does it, at what match rate, and can I see the failures? Match rates are marketing numbers.

05

CHANGE CONTROL

What happens when you change methodology? Do I get notice — and restated history?

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Agenda for Next Class

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Project Ideas & Presentations

Key details, expectations, and submission guidelines for the upcoming week

Ideas Presentations

Come prepared to share your initial project ideas with the class.

Individual vs. Team

Individual: Start refining your project focus.

Teams: Connect this week to **brainstorm and align** on ideas.

Expectations Deck

A separate slide deck with detailed **guidelines & expectations** will be shared in advance.

Please review it carefully before presenting.

Submission

A submission link will be shared on **Canvas** to upload your completed ideas deck.

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Guidelines & Expectations

(Presentation Date: Wednesday, September 16)

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Overview

- **Purpose of the Idea Pitch (5% of grade)**: Present a concise, well-researched project concept

that shows how alternative data creates value.

- **What to Cover:**
  - Clearly define the problem or opportunity.
  - Identify potential datasets and key stakeholders.
  - Outline your initial project plan and approach.
- **Format**: Total time 15–20 minutes, including Q&A and open discussion.

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Evaluation Criteria (Those with * are Mandatory)

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- **Overview***
- **Research on the topic**
  - Industry background and current status (macro and micro), global and US (as relevant)
  - Trends in the Industry and projections for growth/decline, global and US (as relevant)
  - Competitive analysis and profiles or major competitors, global and US *
- **USP of your project idea (**Is it actually an Alternative Data use case? What is the proposed innovation? How do you create alpha?**)***
- **Target Audience/ Customer***
- **Market sizing and market segmentation**
- **Dataset Sourcing and Evaluation Approach***
- **Data Cleaning and Feature Engineering Approach**
- **Use of a PM framework from Week 3 to evaluate your product***

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Tips for a Strong Presentation

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- Keep your presentation clear and visually engaging.
- Back up claims with data and credible sources.
- Clearly highlight your project's differentiators and innovation.
- End with discussion prompts to spark class participation.
- Stay within the 15–20 minute time limit (including Q&A).

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Logistics & Submission

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- **Deadline:** Just before class – 1:40pm on Sept 16th
- A **link will be posted on Canvas** to upload your presentation.
- Review these guidelines carefully before finalizing your presentation.


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Peer feedback using Plause
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- Providing and receiving feedback are two very important things that are a part of this course. For all your presentations we will make use of Plause (https://www.plause.app/), created by our very own **Prof Ryan Bolick**.
- A detailed guide on how to use Plause will be shared soon.

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Thank You & Questions

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Back up

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Another (Dated) Example: Facebook's "M" Assistant (2015–2018)



What Everyone Saw....



Facebook announced **"M"** in August 2015 - an AI assistant that could book restaurants, buy gifts, and plan trips.

Positioned as competition to **Siri and Google** Assistant.

Media excitement about an AI breakthrough.

Demo videos showing complex task completion.

M gave tone-deaf suggestions like "make dinner plans" when friends discussed "exsanguinated corpses" in a book

Human workers were **required to train the AI,** but users didn't know when humans took over

No more than **30% of M's** answers were ever served by the AI system

Over **70%** of requests were answered by human operators

What the PM Actually Dealt With.....

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TECH

Facebook is shutting down M, its personal assistant service that combined humans and AI

PUBLISHED MON, JAN 8 2018•1:10 PM EST

THE VERGE | Casey Newton



What Everyone Saw....

- Cool AI features
- Impressive demos
- Tech headlines

What the PM Actually Dealt With.....

- Convincing stakeholders that AI is worth the investment.
- Managing data scientists who speak a different language.
- Explaining to users why AI sometimes fails.
- Measuring success beyond accuracy metrics.
- Dealing with AI that works in demos but fails in production.

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So, Product looked magical in demos, but in production it collapsed. This is the same trap many data products fall into, they look great on paper, but fail when scaled

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The Five Levels of AI/Data Maturity

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Level 1 – Awareness

- Organization recognizes AI/data potential, but **no active projects**
- Activities: reading reports, attending conferences, exploring vendor pitches
- Resources: no dedicated team, minimal investment
- **Example:** A traditional retailer curious about geolocation data but not yet using it

Level 2 - Active

- Early **pilots and proofs-of-concept** in isolated teams
- Focused on experimentation, not production
- Limited coordination; ROI unclear
- **Example:** A bank testing a pilot with credit card data to predict spending patterns

Level 3 - Operational

- AI/data products are **deployed in production** for specific business functions
- Measurable business impact begins
- Still function-specific, not organization-wide
- **Examples:**

– Hedge fund using credit card transactions for trading models

– Retail chain using geolocation data for new store site selection

Level 4 - Systematic

- AI/data are **integrated across multiple processes**
- Enterprise-level strategy, data governance, cross-functional AI teams
- Scaling adoption across departments
- **Example:** Retailer combining satellite, social, and credit card data for demand forecasting

Level 5 - Transformational

- AI/data **fundamentally reshape the business model**
- Organization becomes “AI-first” or “data-first”
- New revenue streams created from data/AI
- **Example:** Tesla → autonomous driving powered by data + AI